Biodiversity and Conservation Analysis in U.S. National Parks

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Introduction

- The U.S. National Park Service (NPS) plays a pivotal role in the preservation of North American biodiversity.

- Using data-driven analysis, we aim to uncover patterns in species endangerment and determine if certain taxonomic groups face higher risks of extinction.

Objectives

- **Distribution Analysis:** Characterize the conservation status across 5,541 unique species.
- **Risk Assessment**: Identify which types of species (Mammals, Birds, etc.) are most likely to be endangered.
- **Statistical Significance**: Use Chi-Square testing to see if protection status is significantly linked to species category.
- **Observation Patterns**: Analyze over 3.3 million sightings to find the most common species in each park.

Data

species_info.csv - contains data about different species and their conservation status. This dataset contains the following columns:

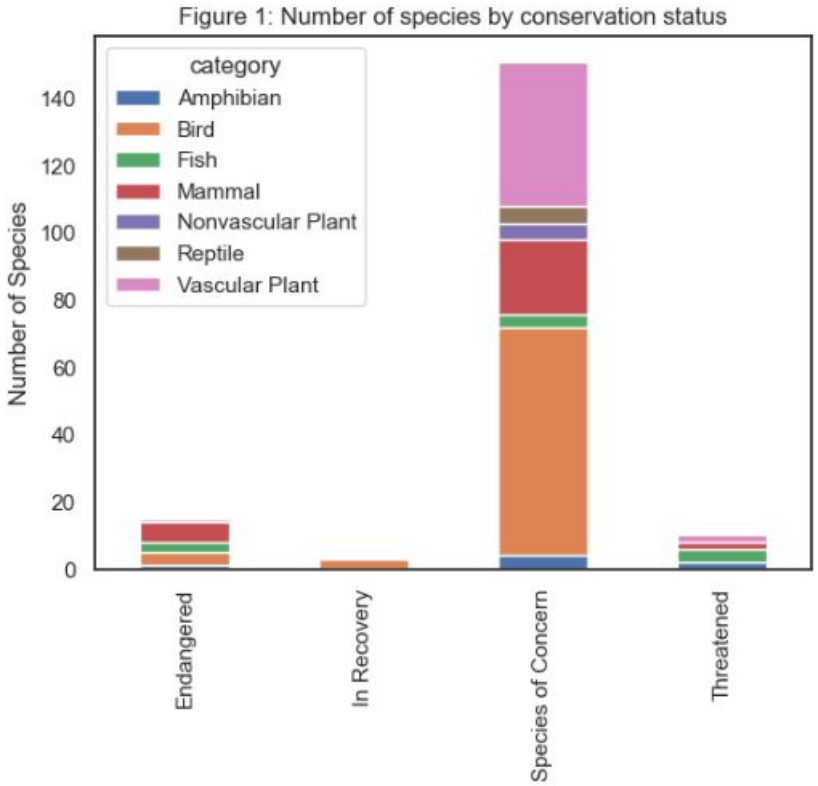
- **category** - class of animal
- **scientific_name** - the scientific name of each species
- **common_name** - the common names of each species
- **conservation_status** - each species' current conservation status

Data

observations.csv - holds recorded sightings of different species at several national parks for the past 7 days. This dataset contains the following columns:

- **scientific_name** - the scientific name of each species
- **park_name** - Park where species were found
- **observations** - the number of times each species was observed at park

Distribution Analysis

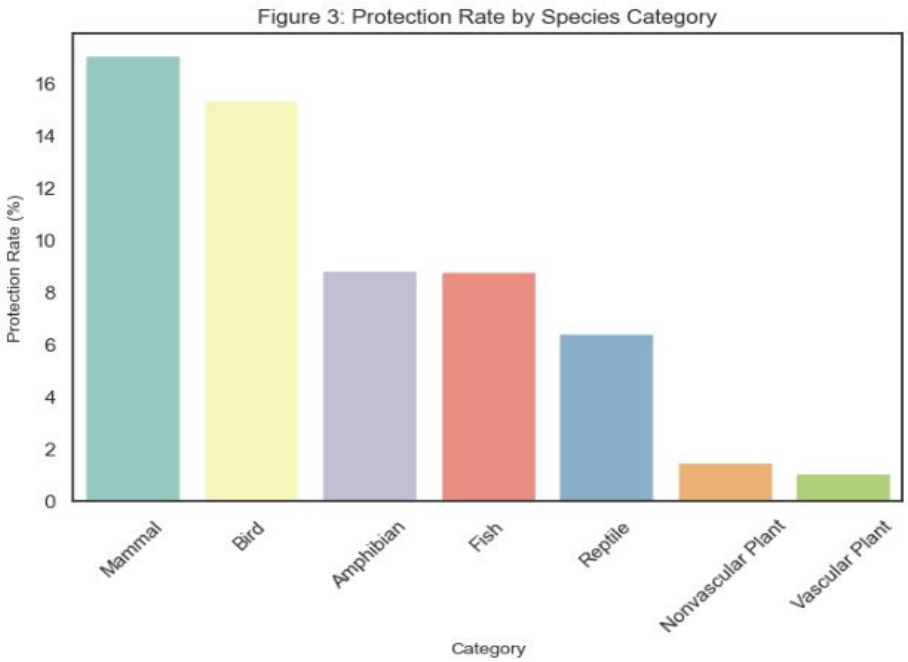


- While the vast majority of species are currently classified as "No Intervention", **3.2% (179 species)** are at some level of risk.

- **Species of Concern:** Birds are the most represented in this category (68 species)

- **Endangered:** Mammals and birds account for the majority of species facing immediate extinction threats.

Risk Assessment



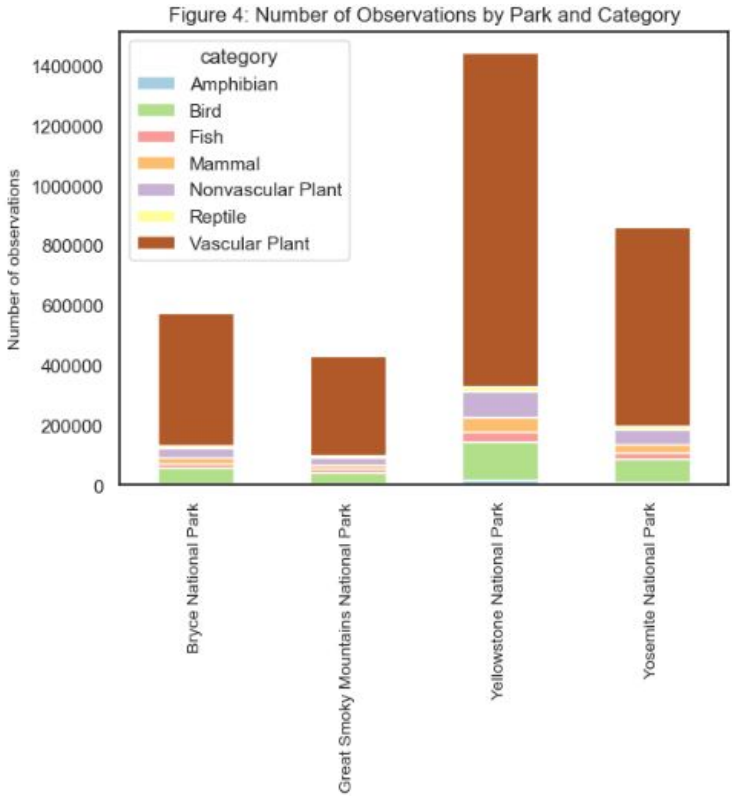
Mammals (17.05%) and **Birds** (15.37%) have significantly higher protection rates compared to Plants (<2%).

Statistical Significance

A **Chi-Square Test of Independence** yielded a p-value < 0.05, confirming that the association between species category and protection status is statistically significant.

Explanation: This would mean that species are not all at the same level of conservation risk, and some categories are significantly more (or less) likely to be protected than others.

Observation Patterns



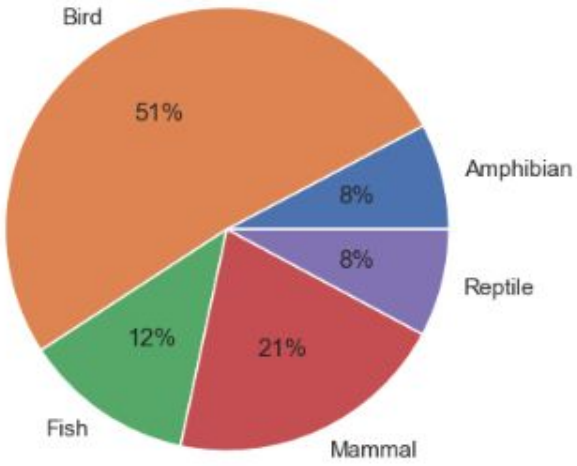
- Biodiversity Balance: Across all parks, plants dominate the landscape, accounting for **83% of all observations**.

- **Yellowstone National Park** recorded the highest activity (1.4M observations).

- While **Bryce** and **Great Smoky Mountains** saw the lowest (under 0.6M)

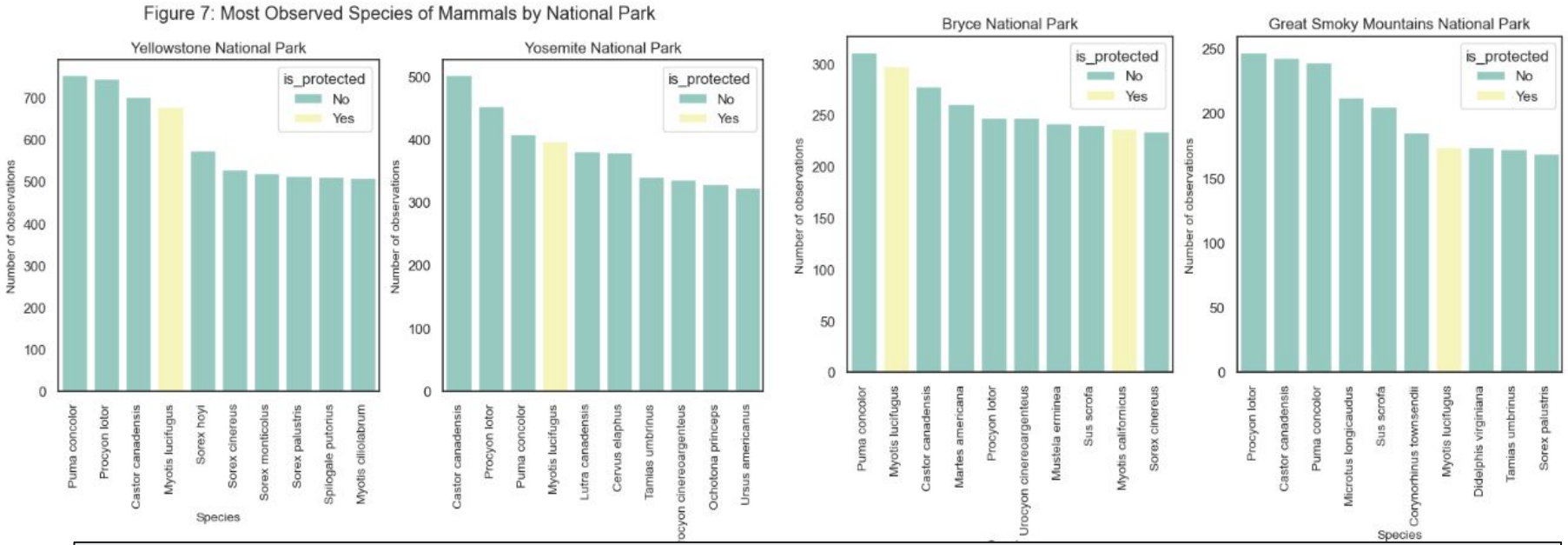
Observation Patterns

Figure 6: Animals - Number of Observations by Category, in percentage of total



- Among animals, **birds** are the most frequently sighted (51%), followed by **mammals** (21%).

Observation Patterns



The **Little Brown Bat** (Myotis lucifugus) stands out as a unique case—it is the only protected mammal that consistently ranks in the "Top 10" most observed mammals across all four parks, indicating a widespread but vulnerable population.

Observation Patterns

- **Common Sightings:** The Puma (Puma concolor), Beaver (Castor canadiensis), and Raccoon (Procyon lotor) are the most ubiquitous mammals across all parks.

Summary

- **Protection Disparity:** Mammals (17.05%) and Birds (15.37%) have significantly higher protection rates compared to Plants (<2%).
- **Statistical Result:** A Chi-Square Test yielded a p-value < 0.05, confirming that the association between species category and protection status is statistically significant.
- **Most Observed:** The Little Brown Bat is a notable outlier—it is both highly observed across all parks and carries a protected conservation status.
- **Regional Leader**: Yellowstone National Park recorded the highest volume of sightings, totaling ~1.4 million observations.